



USER MANUAL

1.8m Low Orbit Satellite Transportable Antenna
WFA180SRX

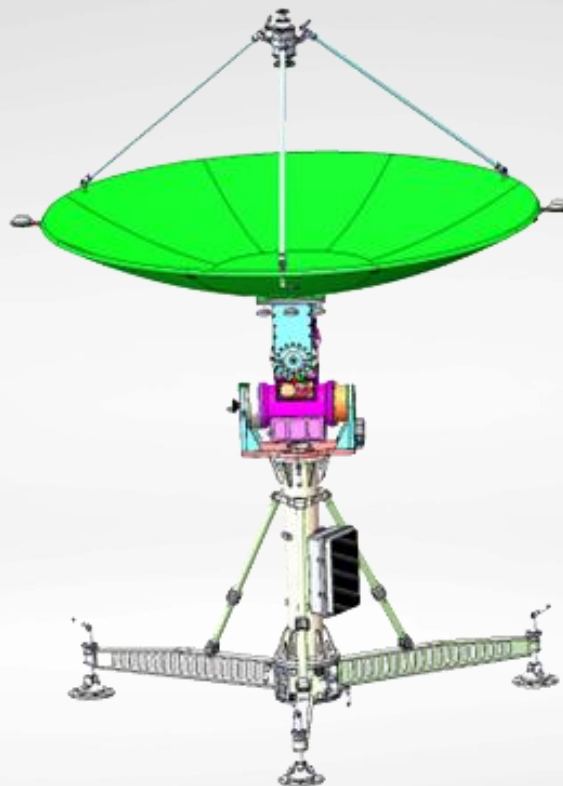


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Revision History

REVISION	DESCRIPTION
A	Initial Release

Agreement

Symbol conventions.

The following symbols appear in this document and their meanings are as follows.

symbol	Explanation of symbols
 Danger	Indicates a hazardous situation with a high or moderate risk which, if not avoided, could result in death or serious injury.
 warn	Indicates a hazardous situation with a low level of risk which, if not avoided, may result in minor or moderate injury.
 Notice	Indicates potential risks which, if not avoided, may result in equipment damage, data loss, equipment performance degradation, or unpredictable results.

Special Notes

Please read carefully all the places marked with attention,  warning, danger and other symbols  in this article! And the product users must strictly abide by the precautions, otherwise it will affect the normal use of the equipment and cause accidents such as personal injury.  **If the user does not operate according to the regulations, resulting in personal injury or equipment failure, our company will not be responsible.**

1. System Overview

The 1.8m Low Orbit Satellite transportable Antenna can quickly receive satellite signals. The antenna has an equivalent aperture of 1.8m and is multifunctional, high-performance, and highly automated.

The antenna system consists of the antenna feed subsystem, structural subsystem, servo control tracking system, and supporting cables. The system composition is shown in the figure below.

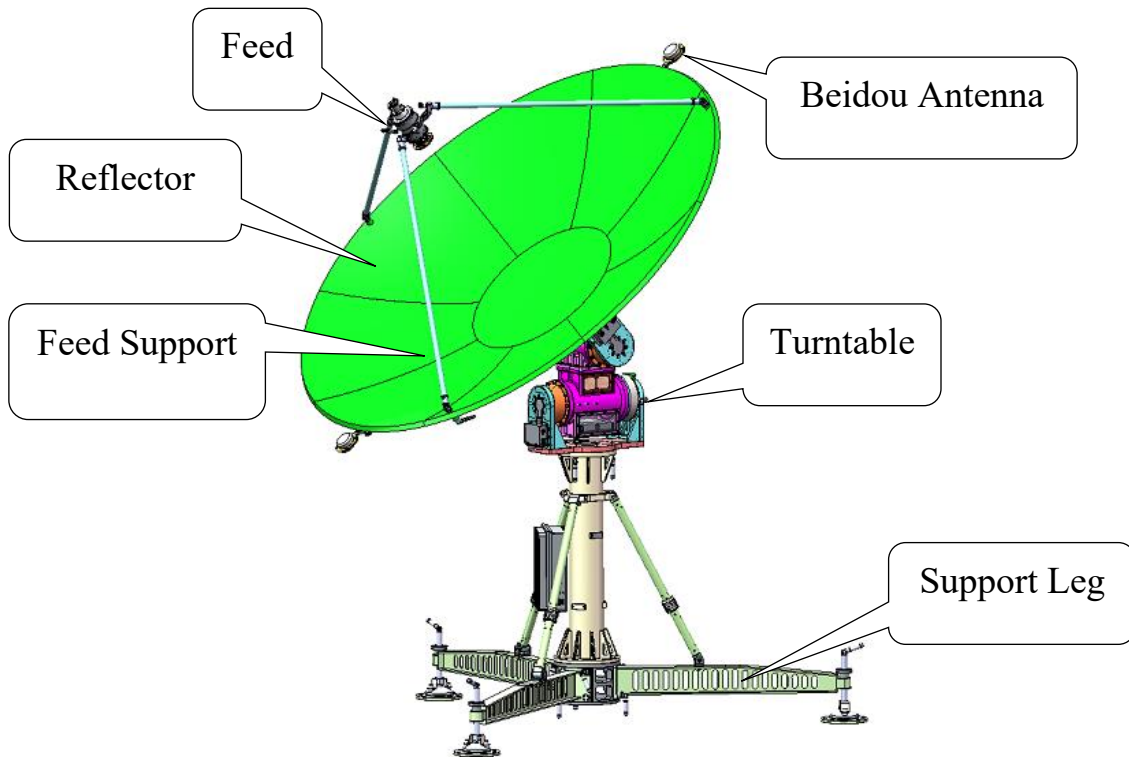


Figure 1.1

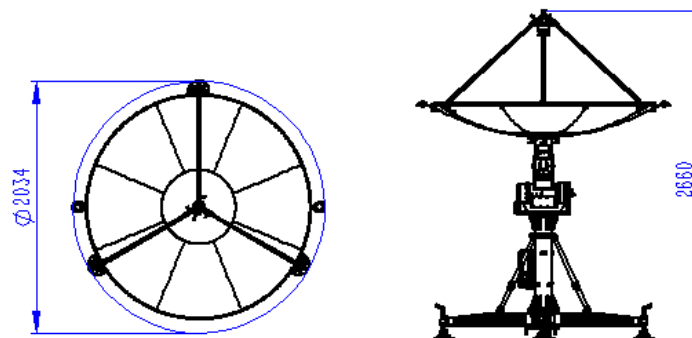


Figure 1.2

2. Pre-installation Preparation

Before starting the installation, make sure you have studied and understood the information in this document.

2.1 Installation Environment Preparation

- 1) The best placement of the antenna should follow
 - a) Horizontal installation: Try to choose a horizontal installation plane;
 - b) Choose a relatively flat site and make sure there are no tall buildings, trees or other obstructions in the direction of the antenna.
 - c) Clean the site and ensure that there are no foreign objects within 1 meter around the antenna to prevent collision when the antenna is rotated;
 - d) Please ensure that there are no large electromagnetic field equipment such as microwave relays and radar stations operating around the antenna operation site.
- 2) Location Avoidance
 - a) Areas that are 10°C higher than the ambient temperature, such as flue openings;
 - b) Environments with heavy smoke and dust, such as flue openings;
 - c) The same horizontal plane as the transmitting antenna for high frequency, ultra-high frequency or radar signals.

2.2 Antenna Dimensions

The installers must have basic safety operation knowledge, master the correct operation methods, and have the corresponding job qualifications. Users should pay attention to the following two points when organizing installers:

1. Installers should master certain installation and erection methods;
2. The number of people for installation is generally 2 to 3.

3. System Installation

3.1 Unpacking and Inspection

Unpack and inspect the goods that arrive at the site to check whether all equipment is complete and intact. The packing list is as follows:

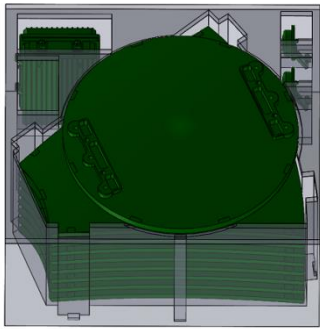
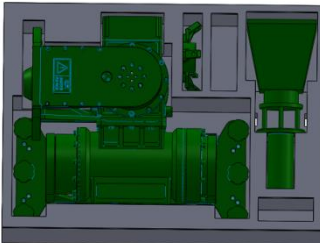
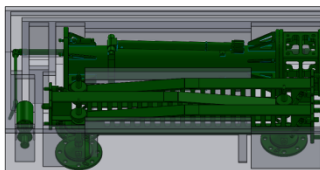
Serial number	Schematic diagram	Parts or functions	quantity	Remark
Box 1		Center plate-packaging Facial flap-packaging BeiDou Power Box	8 petals	Identification number: 1, 2, 3, 4, 5, 6, 7, 8,
Box 2		Turntable L, S antenna feed components Feed fixed	1 set each	/
Box 3		Support legs (3 pieces) Column L, S antenna feed components Feed support rod assembly	1 set each	/

Table 3.1



Pay attention during transportation, handling and installation:

1. Please place the equipment according to the stacking requirements printed on the packaging box to avoid damage caused by inverting the equipment.
2. Avoid collision between the equipment and doors, walls, cargo and other objects.

Procedure

Step 1: Check the completeness of each packaging box

- 1) If the packaging is intact, proceed to **step 2**;
- 2) If the packaging is severely damaged or soaked in water, find out the cause and report it to the company's sales staff.

Step 2: Check the type and quantity of equipment in the box against the packing list

- 1) The type and quantity of items should be consistent with the packing list ;
- 2) If there is a problem with wrong goods, out of stock, or if the goods are found to be damaged, please report it to the company's sales staff in a timely manner.



Note: To protect the equipment and find out the cause of equipment damage later, the packaging boxes and all equipment should be moved indoors and stored properly. Check the equipment storage environment, take photos of rusted or corroded equipment, packaging boxes, and packaging materials, and keep them on file.

3.2 Antenna Installation Steps

After the antenna installation location is selected, start installing the antenna. The antenna installation diagram is as follows:

Step 1: Install the support legs

Open the No. 3 packaging box and take out the column and support leg. Fix the support leg on the column according to the position shown in the figure below, and lock the two locking handles of the support leg. At the same time, lock the pull rod on the support leg and the pull rod on the column with nuts. The installation diagram is shown in the figure below:

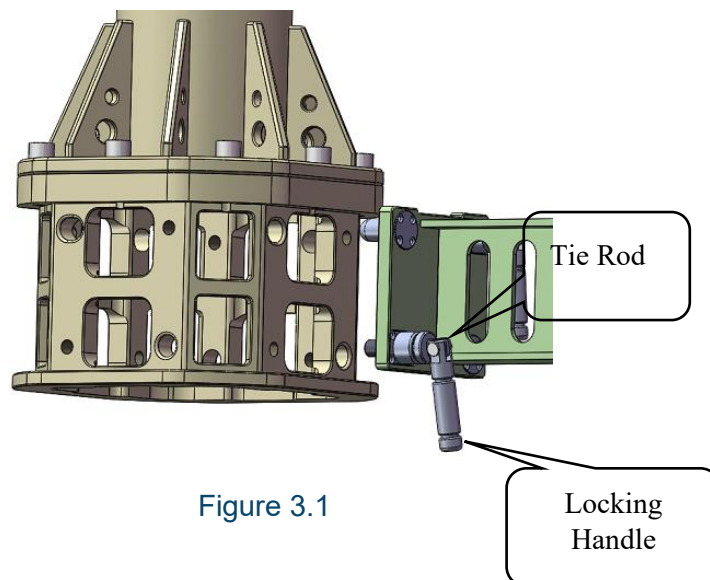


Figure 3.1

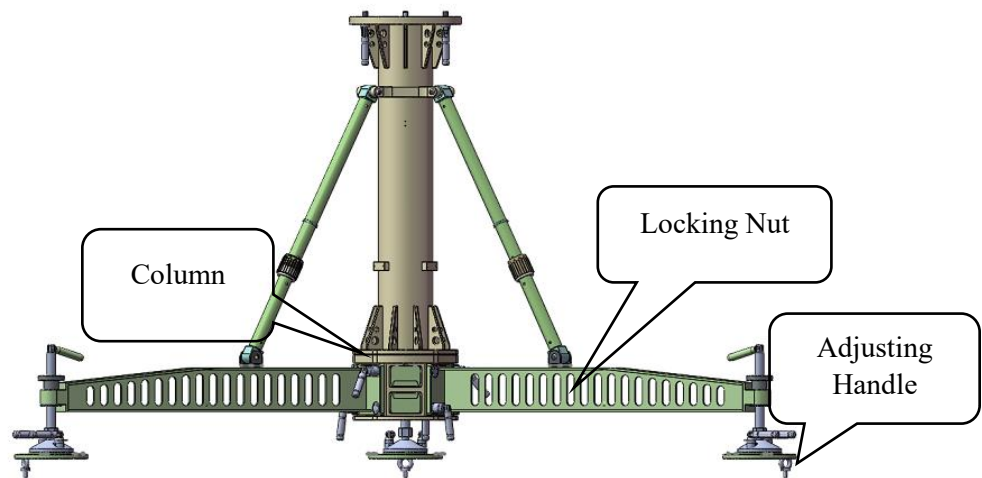


Figure 3.2

The completed status is shown in the figure below;



Figure 3.3



Danger: Before locking the handle, adjust the length of the three legs and roughly adjust the middle column to be level with the ground to prevent the column from being uneven after the turntable is set up, causing the antenna to tilt. (For fine adjustment, see step 5 of 3.2)



Pay attention to the three supporting legs and make sure the handles are locked .

Step 2: Install the turntable

Open the No. 2 packaging box and take out the turntable assembly. Place the turntable on top of the support column, insert the turntable into the positioning hole and lock all the handles of the column . Then power on the antenna and restore the position of the antenna turntable to 0 degrees on the X axis and 0 degrees on the Y axis.

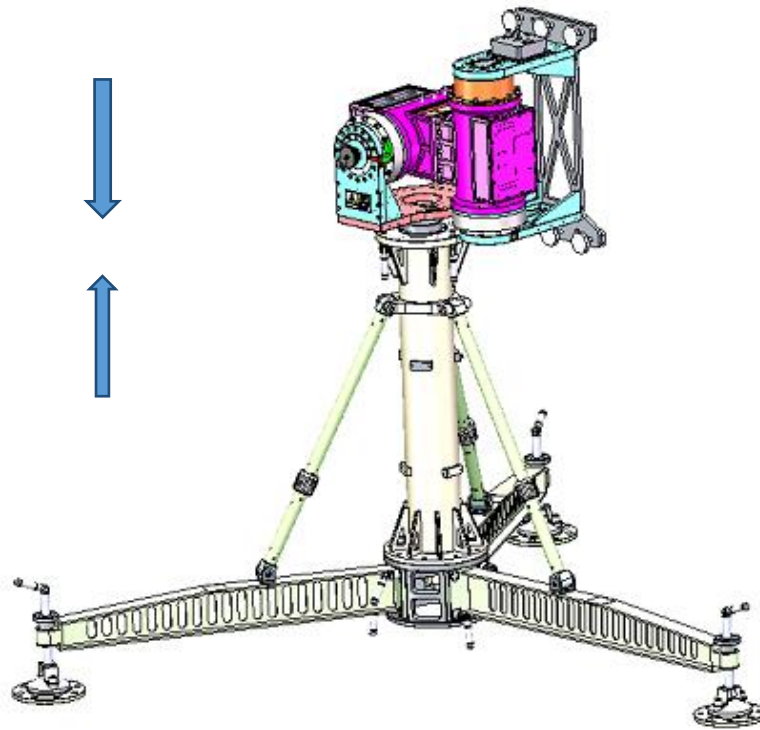


Figure 3.4

The completed status is shown in the figure below:

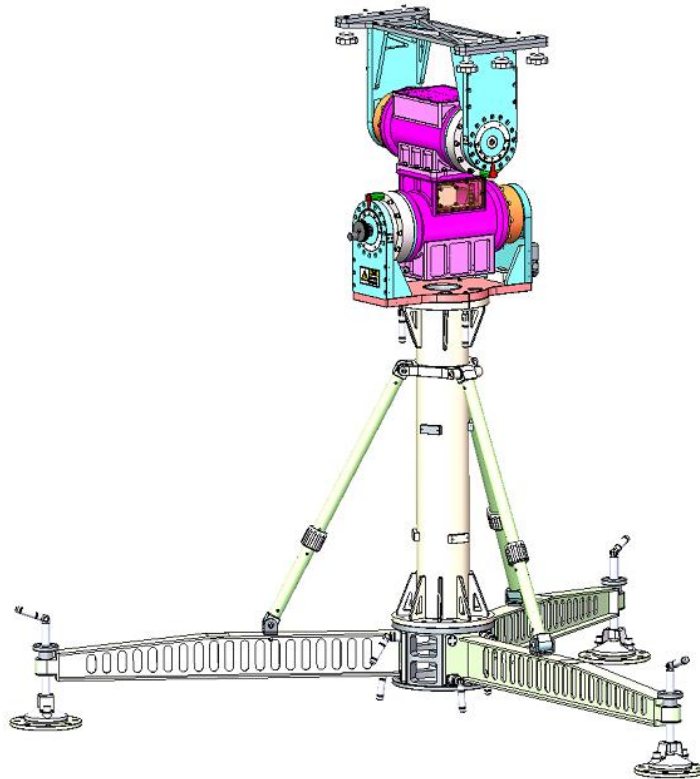


Figure 3.5



Danger After setting up the turntable, make sure the locking handle is tightened.
Detail photo of the locking handle

Step 3: Installation of antenna reflector

1. Open box No. 1 and take out the center disk, place the center disk on the turntable connecting plate, and fix the center disk on the turntable from the reverse side using the captive handle. As shown in the figure below;

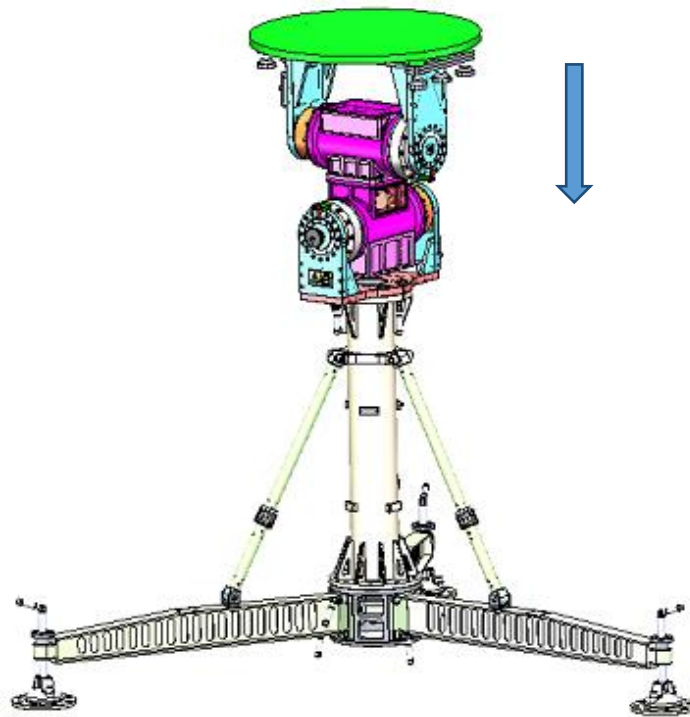


Figure 3.6

2. Open box No. 1 and take out 8 face petals. Install the 8 face petals in the order of 1, 2, 3, 4, 5, 6, 7, and 8, and secure them with the locking handle. The numbers of the buckles should correspond one by one (one by one with the number of the center plate);

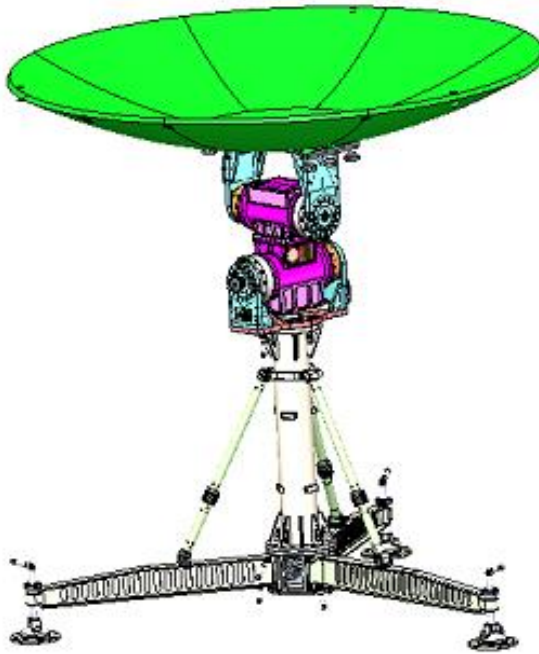


Figure 3.7



sure all the buckles are in place and locked. Add detailed photos



Note that the installation order of the flaps must be in accordance with the marked directions.

3. Dual Beidou installation

There are two Beidou mounting holes on both sides of the antenna surface. After alignment, they can be locked by tightening the captive handle. Note: The installer should face the front of the power switch and install BD1 on the left and BD2 on the right.

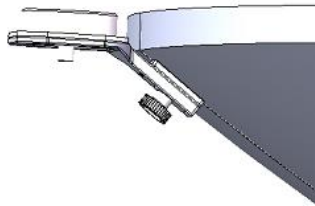


Figure 3.8

4. Pole assembly

Open box No. 1 and take out the feed fixing ring and feed , place the feed at the designated position and lock it with the locking handle (take a photo) , then take out the support rod from box No. 3 and connect the support rod head to the feed fixing ring . After installation, it will be as shown in the figure below.

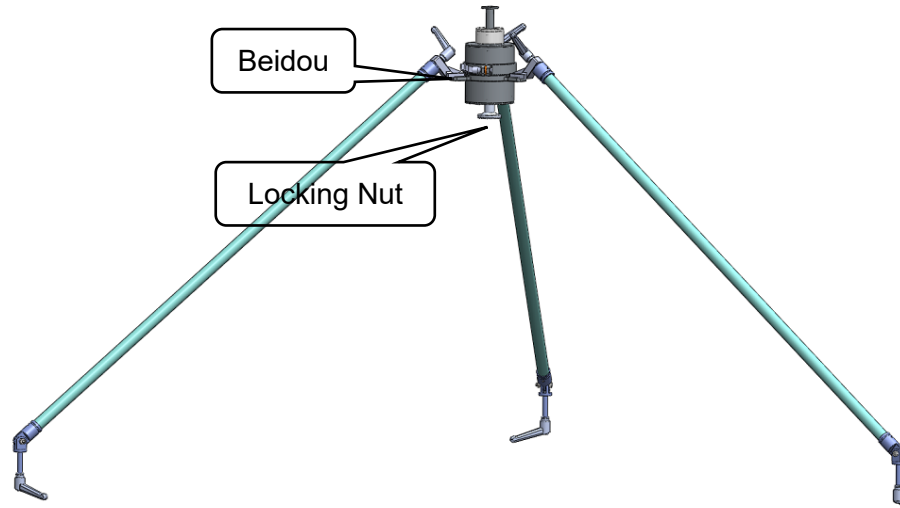


Figure 3.9

5. Feed installation

Pass the installed support pole assembly through the pin hole on the antenna surface to secure it, and then install the support pole assembly through the fixing handle.

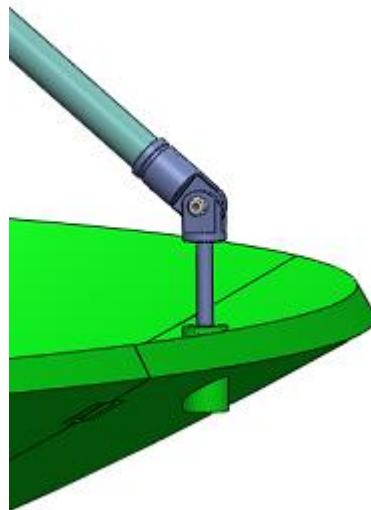


Figure 3.10

The completed status is shown in the figure below:

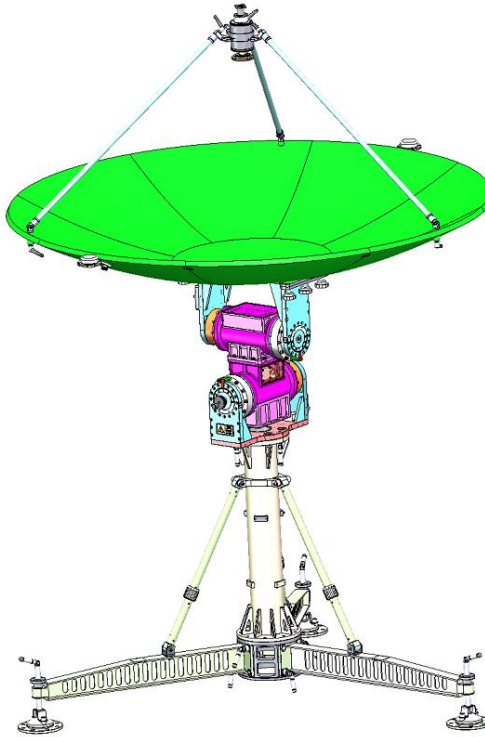


Figure 3.11

Step 4: Feed replacement

Feed replacement requires two people. First, move the antenna to 0° on the X axis and 90° on the Y axis (take photos & videos) , then loosen the fixed handle at the feed, remove the feed and place it in packing box No. 2, open packing box No. 3 and take out the feed components of other frequency bands, take out the feeds of other frequency bands and place them in the fixed designated position of the feed, then lock it with the fixed handle. Feed replacement is complete.

Step 5: Level adjustment

Special note: After the antenna is erected, it must be adjusted horizontally. A high-precision digital level is provided with the antenna. Place it on the turntable horizontally on the ground, and adjust the knobs of the four legs to make the digital display to 0 degrees (the allowable error is plus or minus 0.1 degrees).

3.3 Antenna Disassembly and Packaging

The disassembly and packaging process of each module of the 1.8m antenna is the opposite of the installation process.

3.4 Cable Connection and Safety Check

3.4.1 Wiring Instructions (Change Diagram)

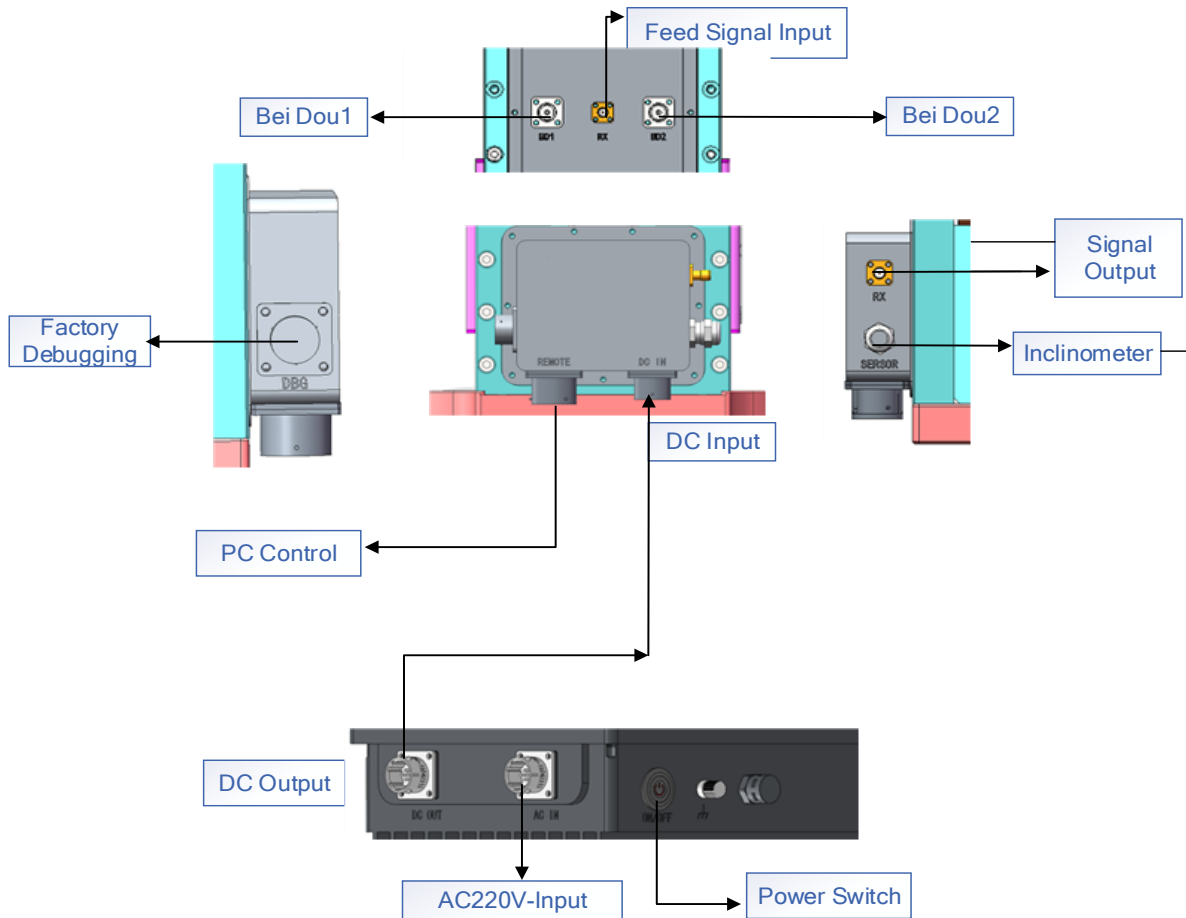


Figure 3.12

- 1) As shown in the figure above, connect the cables one by one;
- 2) Connect the power cable marked with AC220V to the AC IN of the antenna power box ;
- 3) Connect the power cable with DC OUT mark to the DC IN of the antenna turntable ;
- 4) The dual Beidou RF cables are connected to the antenna surface and both sides of the turntable, and the corresponding interfaces are BD1 and BD2;
- 5) Use an Ethernet cable to connect the antenna turntable REMOTE to the PC end.



Dangerous antennas are powered by AC power. Cables and power supplies with AC power must be disconnected from the main power before plugging or unplugging. The switch button should be in the off state.



Danger Before powering on each time, carefully check whether the AC cable is damaged. Do not use it if it is damaged.



Warning: When connecting any cables on the antenna, the antenna must be powered off. Any power-on operation is prohibited!

3.4.2 Security Check

After the antenna system and related equipment are all installed, it is necessary to check the installation items and installation environment, and check whether the cable-related items are correct.

The following table shows the items that need to be checked after the system is installed:

Serial number	Inspection items
1	Check whether the antenna bracket is leveled and installed firmly.
2	Is the antenna turntable installed firmly?
3	All cables are connected correctly.
4	Check whether the connector plug is firmly connected.
5	Is the bending radius of the cable greater than 200 mm
6	Check whether the connection handles are tightened and the antenna flap buckles are in place.
7	Check whether the system power supply is stable
8	Make sure the cables are securely fastened to prevent them from getting tangled when the antenna is rotated.

Table 3.2

3.5 Software Control

The host computer software control instructions are general instructions. Please refer to the host computer software to actually control the antenna movement, etc.

3.5.1 Software Support System

Software support systems include:

system	Support	Install Dependencies
Windows 7	support	.NET 47 and SP1
windowSever2000	support	.NET47
Windows 8 and above	support	unnecessary

Table 3.3

The installation method of .NET47 is: download the .NET47 package from the Internet and install it. Or contact the administrator.

Software support minimum hardware requirements:

- Hard disk 50GB
- Memory 8GB
- CPU:i3-5000
- Graphics card: Integrated

If you need to run: The minimum requirements for the 3D Data Interface are as follows:

- Hard disk 150GB
- Memory 16GB
- CPU:i3-11000
- Graphics: [9500 GT](#) 2GB

3.5.2 File System Introduction

The software files consist of the following:

File name	Describe	Details
config	For database file storage	Among them, "system.db" is the system file. The others are antenna and satellite database files.
ConfigTime	Used to store satellite 2 row elements	Each satellite type is named, and the 2-line root number information of the imported satellite is stored internally. You can view the 2-line root number information of the historical import.
Pack	For package management	The software supports package function management and antenna control templates can be added through packages.
Resource	For resource management	This file cannot be modified. It is used to store external antenna logic, external web control, 3D resources, etc.
Gmaptx.exe	Program startup file	

Table 3.4

When using software version "XY-24.4.X" series or below

The following prompt appears when you run the software for the first time: "Please contact the administrator to authorize the current user first."

Place the XY file in "C:\Users\Administrator\AppData\Roaming" and replace it if it already exists. Administrator = the administrator path.

When using the software version "XY-25.XX" series:

The following prompt appears when you run the software for the first time:

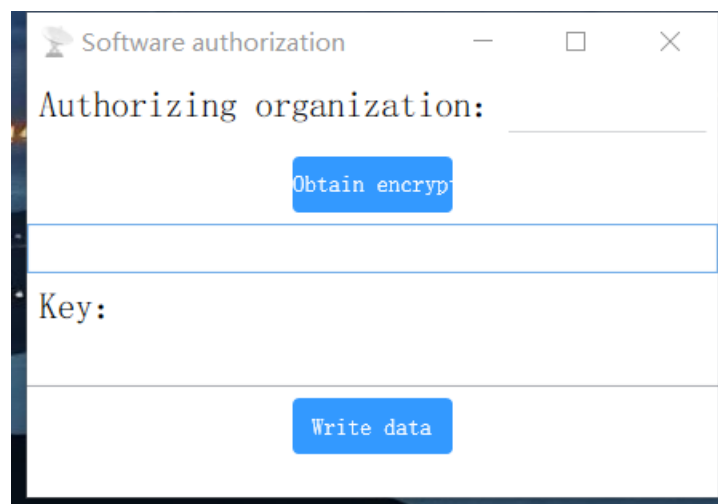


Figure 3.13

- Authorized unit: software user,
- Key: Authorization password.
- First, fill in the authorization unit to obtain the encryption password, send the encryption password to the administrator, the administrator will give the key, write the key to the file, and it's done. Start the software again.

After the authorization is completed, the software can be used permanently on this computer.

3.5.3 Creating the Antenna

"XY Test Antenna " in the "Create Antenna Name" position ; the antenna name cannot be empty, and the antenna name must be unique, and select the antenna type. After the antenna is completed, click the antenna name to enter the "XY Test Antenna" to get the interface shown in the figure.

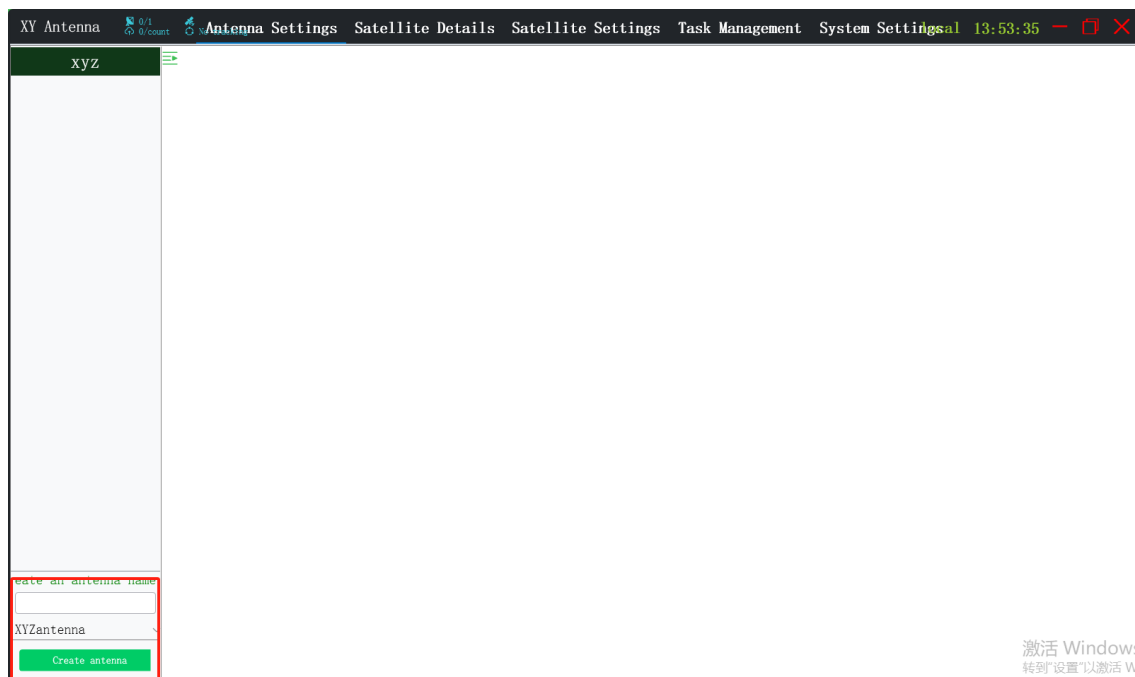


Figure 3.14

1. Antenna connection: The antenna connection methods are divided into TCP and serial port connection. The default TCP connection of the antenna is generally: "192.168.100.7" and the port is "23". The antenna connection entrance is slightly different for different antenna types. Click the button on the right of the antenna name to enter the antenna interface, select Settings in the antenna control interface, and set the antenna connection method in Settings.
2. After clicking the start button and connecting successfully, the interface changes from a prohibited state to a controllable state.
3. Time acquisition: Click the yellow-green "Local 09:14:17" text, and the time will switch to antenna 09:14:18. Antenna: The time must be used after the antenna is connected.

4. You can see whether the antenna is successfully started by looking at the antenna icon on the left side of the software header. If the antenna is successfully started, the page will change to 1/1.
5. Antenna deletion: Select the antenna name on the right, right-click and select Delete to delete the antenna.

3.5.3.1 Status Information

The first time the software controls the antenna, it must execute "Collection" until the collection is completed.

The antenna's motion states generally include collection, XY speed operation, XY pointing operation, azimuth and elevation pointing, standby, etc. (Screenshot)

Location information: longitude, latitude, altitude (km). (Note: Displays the current system usage values)

Attitude information: Heading indicates the degree of north deviation, elevation indicates the degree of horizontal roll, and roll indicates the degree of horizontal tilt. (Note: Displays the current system usage value)

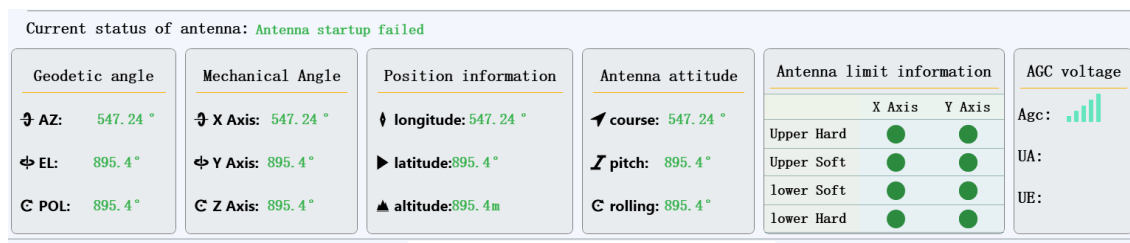


Figure 3.15

3.5.3.2 Status Settings

- Attitude information: Attitude information is divided into manual attitude and system attitude. System attitude is the horizontal Beidou data of the antenna. Manual attitude is the manually set heading, elevation and roll. When using manual attitude, there are two ways to adjust. Method 1: Put the antenna in the stowed state, place the level under the antenna surface, and make the values of the level in each direction zero. At this time, select the manual attitude and fill in the elevation and roll as zero degrees. Then adjust the heading. Method 2: Put the antenna in the stowed state, first select the system attitude, refer to the current system elevation and roll and fill them in the attitude information, and then set it to manual attitude. Adjustment of heading: Adjust the attitude by pointing to the satellite, such as: pointing to "110.5" the current theoretical heading is: 152.14, 57.14. The actual maximum value signal is: 152.54, 47.15, and the current manual heading is: 187.41. The heading calculation formula is: current heading + (theoretical value - actual value) = 187.41 + 152.14 - 152.54 = 187.01. Fill in "178.01" in the heading.
- Antenna position: The antenna position is divided into manual position and system position. The system position is the Beidou data of the antenna. Manual position: manually set the antenna position information. Note that the antenna height unit is KM. Both longitude and latitude are degrees, such as: 50°30 minutes is modified to: 50.50°.
- Linear compensation angle: X compensation angle is used to compensate for the inaccurate angle in the X rotation direction. Y compensation angle is used to compensate for the inaccurate angle in the Y rotation direction. Linear compensation is generally an antenna system level parameter. Do not fill in the value unless necessary.

- Earth supplementary angle: Azimuth supplementary angle compensates for inaccurate rotation in azimuth, and elevation supplementary angle compensates for inaccurate elevation. (Note: For XY base antennas, the current supplementary angle setting is generally not used)
- Geostationary orbit pointing: Control the earth through the orbiting satellite. Just enter the longitude. The antenna moves to the target position and points to the satellite. When pointing to the satellite, the satellite will have theoretical azimuth and elevation polarization. For left and right polarized antennas, ignore this parameter. When using linear polarization, the calculation method is: actual polarization of the antenna = theoretical polarization of the antenna - mechanical polarization of the antenna. The general polarization value range is: -90~+90. The conversion method is +/-180°

The screenshot shows a web-based control interface for an antenna. At the top, there are three tabs: 'Control', 'Settings' (which is active), and 'Interface Protocols'. Below the tabs, there are several sections of settings:

- Location setting:** Includes input fields for 'longitude: 0', 'latitude: 0', and 'altitude: 0'. It also has radio buttons for 'system' (selected) and 'user', and a 'Settings' button with a download icon.
- Attitude setting:** Includes input fields for 'course: 0', 'pitch: 0', and 'rolling: 0'. It also has radio buttons for 'system' (selected) and 'user', and a 'Settings' button with a download icon.
- Angle compensation:** Includes input fields for 'X Axis: 0', 'Y Axis: 0', and 'Z Axis: 0', and a 'Settings' button with a download icon.
- Satellite Pointing Settings:** Includes an input field for 'SATELLITE LONGITUDE: 0' and a 'Settings' button with a download icon.
- Pointing at the Sun:** A section with a 'start' button.

Figure 3.16

3.5.4 Antenna Control

For the control of XY antenna, the azimuth and elevation are converted into XY movement. The range of movement is: azimuth is 0-360 unlimited rotation, and elevation is: 0-90°. For azimuth and elevation antennas, it is divided into two types: azimuth unlimited rotation and azimuth cannot be infinitely rotated. Mechanical azimuth is not equal to the azimuth of the earth, and mechanical elevation is not equal to the elevation of the earth. When the wireless rotating azimuth antenna is not used, the azimuth range of the antenna operation should be checked on the radar chart. Check whether it passes through the azimuth limit axis. Like other AET type antenna movement and azimuth, when controlling the circular multi-faceted phased array antenna, try to use one face to cover the signal range when tracking the satellite.

- AZ-EL pointing mode: Azimuth (AZ) is the geodetic azimuth and the degree of northing. Elevation (EL): is the geodetic elevation and the degree of horizontal. The AZ degree range is [0° ~ 360°), including 0 degrees and excluding 360 degrees. The EL degree range is [0° ~ 90°]. Enter the value in the respective data box, click Execute, and the antenna will move the geodetic coordinates to the target position.
- XY pointing mode: The XY setting range is [-90° ~ +90°]. Click Execute and the antenna linear coordinates will move to the set degree position.
- Standby (Stop): Will terminate the antenna movement command.
- Collection: The antenna will move to the skyward position.
- North: The antenna will move to the true north direction.
- AZ-EL jog: The up and down arrows indicate the positive and negative superimposed motion in elevation, and the left and right arrows indicate the positive and negative superimposed motion

in azimuth.

- Track Sun: The antenna will point to the current sun direction and track it.

The screenshot shows a web-based control interface for an antenna. It has four main panels:

- SPEED CONTROL:** Contains 'X unit' and 'Y unit' both set to 1, and a four-way directional pad (up, down, left, right arrows).
- Linear position mode:** Contains 'X Axis position' and 'Y Axis position' both set to 0, and an 'execute' button.
- Position mode:** Contains 'Azimuth position', 'Elevation position', and 'Polarization position' all set to 0, and an 'execute' button.
- Position control:** Contains four buttons: 'Stow' (blue), 'Stop' (orange), 'Pointing North' (blue), and 'Step tracking' (blue).

At the top, there are tabs for 'Control' (selected), 'Settings', and 'Interface Protocols'.

Figure 3.17

3.5.4.1 How to calibrate the Antenna

After a basic understanding of antennas, we will install and place the antennas properly to achieve the best performance. The best placement of the antenna is when the antenna is horizontal and the antenna is "pointing north" to completely coincide with the true north. The following will introduce how to complete it through software cooperation.

When creating an antenna as an "XYZ" antenna and using the system attitude, the attitude correction step can be ignored.

Serial number	Step	Operate
1	Connect to the antenna	Select "TCP mode" in "Listen connection" and enter "IP address:" "192.168.100.7" "Port:" "23". Switch between Run/Stop and Start. The 'Antenna name' dialog box has the following fields and options: - Link method: ☒ TCPmode, ☐ UDPmode, ☐ Serial port mode - Serial port: 请选择... (dropdown), Baud rate: 请选择... (dropdown) - IP address: 192.168.100.7, port: 23 - Loop access: ☐ Proactive escalation/iteration - Buttons: Modify (blue), cancel (orange)
2	Manual Adjustment	Manually adjust the antenna base so that the level bubble on the antenna base is in the center or use an electronic level to level the antenna.
3	Adjust antenna azimuth and elevation	Click Collect to obtain the prepared Beidou coordinate information. Automatically point to the synchronous satellite, record the theoretical position, adjust to the maximum signal, record the current position, subtract the previous and next coordinates, compensate the azimuth difference to the heading, and

		compensate the elevation difference to the elevation offset angle during tracking; (If the initial pointing is the maximum signal, no compensation is required.)
4	Manual Setup	Set the compensated heading value to the manual heading and select manual setting. Attitude setting: course: 0 pitch: 0 rolling: 0
5	Check alignment	Set the longitude of other geosynchronous orbit satellites in the alignment and confirm whether the signal is the maximum value previously aligned; complete the antenna calibration.

Table3.5

3.5.5 Satellite Tracking

3.5.5.1 Satellite Information Import

The screenshot displays the 'Satellite Information Import' interface. At the top, there are buttons for 'Website Update', 'Manually import TLE file', and 'Create orbital elements'. Below these, the 'Satellite Preview' section shows 'Input_O3B_6.24' and 'Satellite Preview' button. The 'Satellite screening' section includes filters for 'longitude 0', 'range 0', and a 'Search' button. A table lists satellites with columns for 'SerialNo', 'Name', 'center frequency', and 'Trackable'. The table shows various O3B satellites, with O3B FM6 highlighted. To the right, the 'O3B FM6' details are shown, including 'Trackable' status, 'Remove Satellite' button, and various parameters like 'center frequency: 0', 'Satellite position', 'Satellite type', 'Group name', 'Satellite forecast', 'TLE Data', and 'physical property'.

Figure 3.18

Satellite ephemeris can be updated through the network or manually imported from a file. When manually imported, the file name to be updated in the file is: input_file name, or directly import the two-line root number file of the satellite to be tracked, or create a 6-root number method.

Satellite Preview: Select a satellite category and click Satellite Preview. The current ephemeris will appear in the ephemeris filter table.

Create 6 roots: When creating 6 roots, enter the value parameters and pay attention to the parameter name and parameter value.

Network update: To update the network, you need to add a template in "System Settings", "Add Satellite", "Satellite Name", and the URL is a 2-line and a half root address. The 2-line and a half root URL is: <https://celestrak.org/NORAD/elements/>

Add satellite:

Satellite name: _____ URL: _____ add

Remove satellite: _____ remove ☐ Remove database

Figure 3.19

3.5.5.2 Satellite Information Preview

Select the satellite name on the left and the satellite details will appear, including the satellite's current location information, the number of satellite 2-line elements, the number of satellite 6-line elements, and other information.

Satellite Preview: Input_03B.6.24
Satellite Preview
Satellite trajectory

Satellite screening: longitude 0 range 0
Search

☐ group ☐ priority ☐ Inclination angle of orbit ☐ Sequence Number ☐ Number of roots ☐ 2number of roots

Satellite search Name: 请选择...
Search

SerialNu	Name	center frequency	Trackable
1	O3B FM5	0	☒
2	O3B FM4	0	☒
3	O3B FM2	0	☒
4	O3B FM3	0	☒
5	O3B FM7	0	☒
6	O3B FM6	0	☒
7	O3B FM8	0	☒
8	O3B FM10	0	☒
9	O3B FM11	0	☒
10	O3B FM12	0	☒
11	O3B FM9	0	☒
12	O3B FM15	0	☒
13	O3B FM16	0	☒
14	O3B FM14	0	☒
15	O3B FM13	0	☒
16	O3B FM20	0	☒
17	O3B FM19	0	☒
18	O3B FM17	0	☒
19	O3B FM18	0	☒
20	O3B MPOWER F1	0	☒
21	O3B MPOWER F2	0	☒
22	O3B MPOWER F4	0	☒
23	O3B MPOWER F3	0	☒

O3B FM6
☒ Trackable Remove Satellite

1 40081U 14038C 25174.27543780 -.00000017 00000+0 00000+0 0 9991
2 40081 0.0510 6.1462 0002793 84.3125 269.5727 5.00115590199872
Update time of ephemeris: 2025/6/23 22:36:37

center frequency: 0 MHzModify

Satellite position: longitude: -48.219934150899° latitude: -0.0416077289866863°

Satellite type: 圆形轨道卫星 赤道轨道 中轨道卫星

Group name: 请选择... Modify priority: 0

Satellite forecast: Observation duration 12hour starting time

TLE Data:

Satellite name : O3B FM6
Launch object number : C
BSTAR drag coefficient : 0
Inclination angle of orbit : 0.051
Number of Orbits Per Day : 5.001155901

Satellite ID : 40081
Time () : 25 年份
orbit model : 0
: 6.1462 升交点赤经
Number of Orbits Since Launch : 99872

physical property:

semi-major axis (a) : 14417.722 KM

Figure 3.20

Satellite track preview: will show the current satellite location information. Select in the ephemeris filter table, and the current satellite track operation diagram will appear on the flat 2D map.

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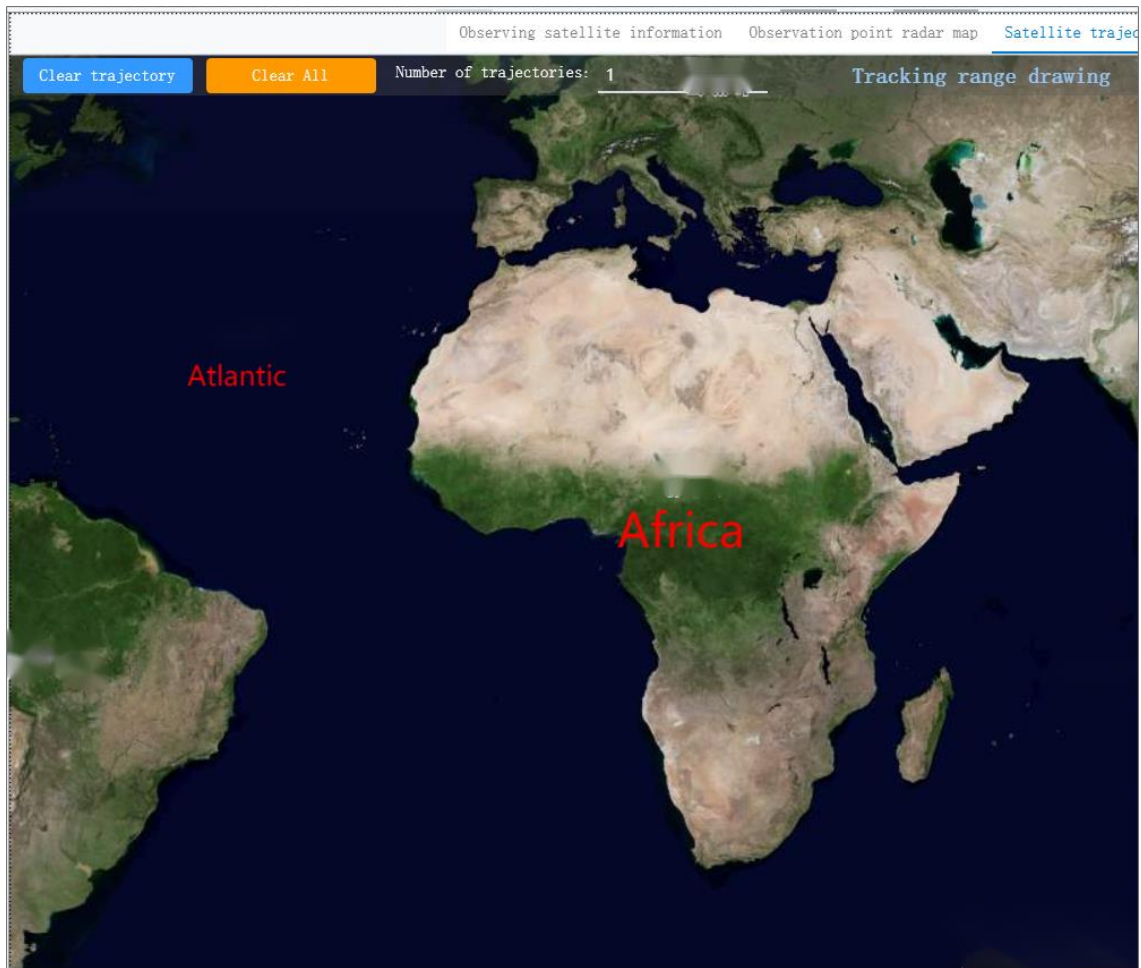


Figure 3.21

Physical properties: Parameters derived through 2 and a half rows.

3.5.5.3 Satellite Tracking

Search

Satellite category: Input_03B.6.24

Observation point: 请选择...

Name: longitude: 0 latitude 0 altitude(00): memo: Restricted

Observation mode

Time mode: 55/5

Refresh: 55/5

Satellite search

Name: 请选择...

Search

SerialNumber	Name	End Time	Maximum Elevation	Angle	Observable duration
1	O3B FM2	14:15:20	13.896		13.867
2	O3B FM7	15:03:20	85.069		61.867
3	O3B FM6	15:54:00	89.777		112.533
4	O3B FM9	14:16:56	15.717		15.467
5	O3B FM13	14:40:24	47.082		38.933
6	O3B FM19	14:40:24	47.073		38.933
7	O3B FM18	15:30:32	89.751		89.067
8	O3B MPOWER F2	15:49:12	89.920		107.733
9	O3B MPOWER F3	14:45:44	55.512		44.267
10					

Observing satellite information

Observation point radar map

Satellite trajectory map

Name

group id: Unallocated group

Modify

priority:

Sequence Number:0

1 40081U 14038C 25174.27543780 -.00000017 00000+0 00000+0 0 9991

2 40081 0.0510 6.1462 0002793 84.3125 269.5727 5.00115590199872

Update time of ephemeris:2025/06/23 22:36:37

Cycle Time:287.93minute

Inclination angle of orbit: 0.051

center frequency: 0

MHz

Modify

entry time

the best time

DEPARTURE TIME

The remaining trackable duration of the current cycle

Average tracking speed:Az(方位):0.13 (度/分钟) El(俯仰):1.53 (度/minute).

Current tracking status

Select tracking antenna: 请选择...

Start tracking

Preprocess tracking

Output trajectory calculation duration: 20minute

file name:

Output azimuth and

Output longitude

Earth-satellite angle

SerialNu	Time	AZ	EL	POL	longitude	latitude	distance (KM)
1	27 14:01:29	269.945	19.417	-1.018	-45.974	-0.040	11012.869
2	27 14:01:30	269.945	19.436	-0.973	-45.958	-0.040	11011.119
3	27 14:01:31	269.945	19.456	-0.928	-45.941	-0.040	11009.370
4	27 14:01:32	269.945	19.476	-0.884	-45.924	-0.040	11007.620
5	27 14:01:33	269.945	19.496	-0.840	-45.908	-0.040	11005.871
6	27 14:01:34	269.945	19.516	-0.797	-45.891	-0.040	11004.122

Figure Error! No text of specified style in document.3.22

Satellite type: Select the satellite to be tracked. Observation point: If you have an antenna, select the antenna; if you don't have an antenna, select the satellite. Name: Select only in custom situations. Longitude, latitude, and altitude are only filled in in custom situations.

Click Search, and the trackable target information will be generated on the left. Right-click and select Track. The antenna will start tracking.

Satellite radar map: The radar map displays trackable satellite trajectory icons. Right-click the satellite icon to manually track the satellite.

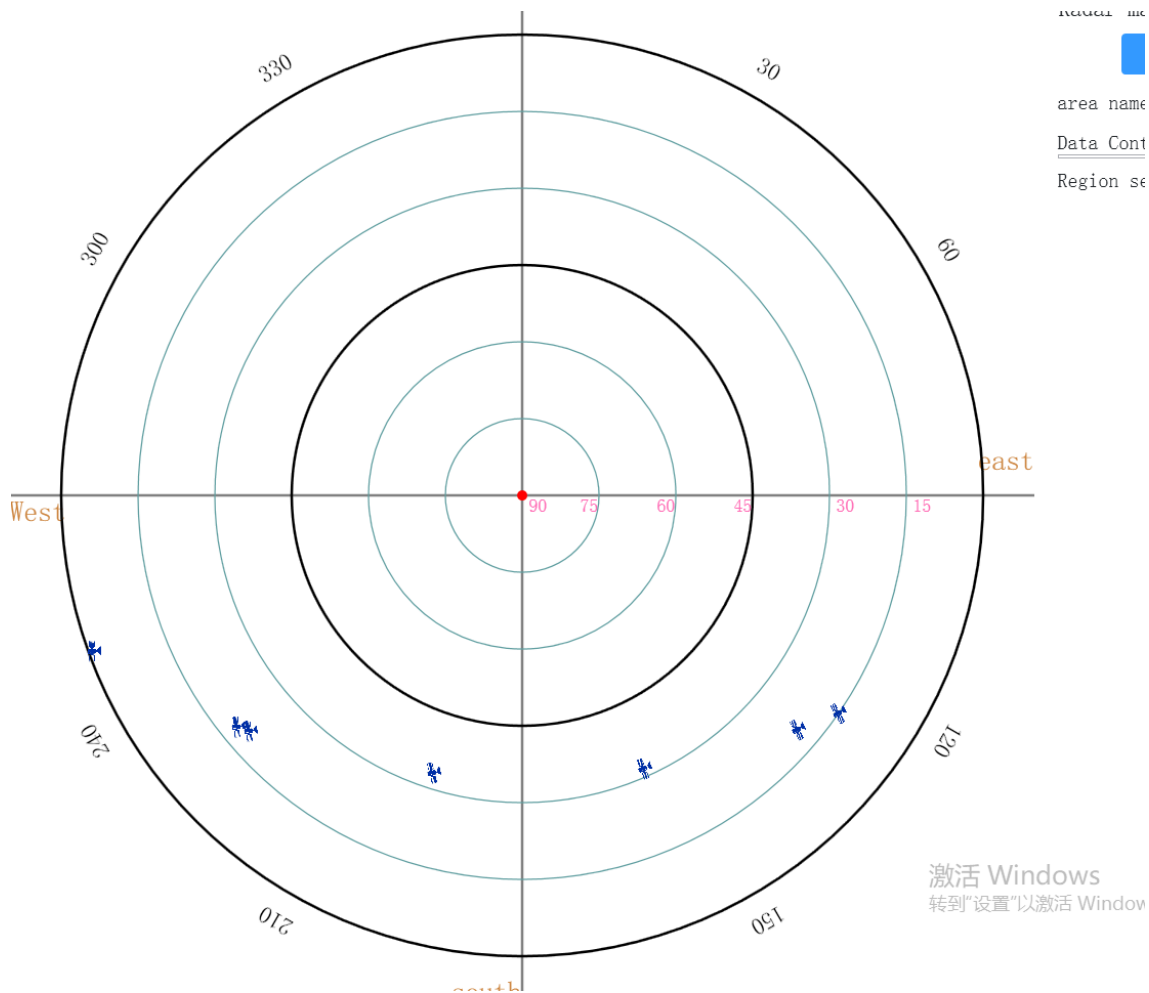


Figure 3.23

Tracking methods are divided into:

- Specified tracking: Select the specified target to track by mouse.

After the calculation is completed, select the trackable target with the mouse and enter the specified tracking.

- Automatic tracking: Set the tracking parameters and select the tracking target in the software for automatic tracking.

Tracking range: Maximum elevation minimum value, the maximum elevation degree that can be tracked within the trackable range, minimum value: cannot be lower than the minimum value. Cannot be greater than the maximum value.

Tracking elevation angle limit: starting minimum degree, ending minimum degree; when started tracking, the current elevation angle is greater than the starting angle. During the movement, when it is greater than the ending angle, the tracking is stopped.

- Preprocessing tracking: Arrange and track future trackable targets.

On the "Satellite Settings" page, change the observation mode to future time and click Find.

Select satellites and perform pre-processing and tracking in "Observe Satellite Information". Mainly used for task scheduling and task-based tracking processing.

- Radar chart area tracking: Set up tracking in the radar chart area.

Draw the tracking radar map range in the radar area, save and start it. It is mainly used for regional signal tracking. When using this function, you have a deep understanding of the satellite signals in this area.

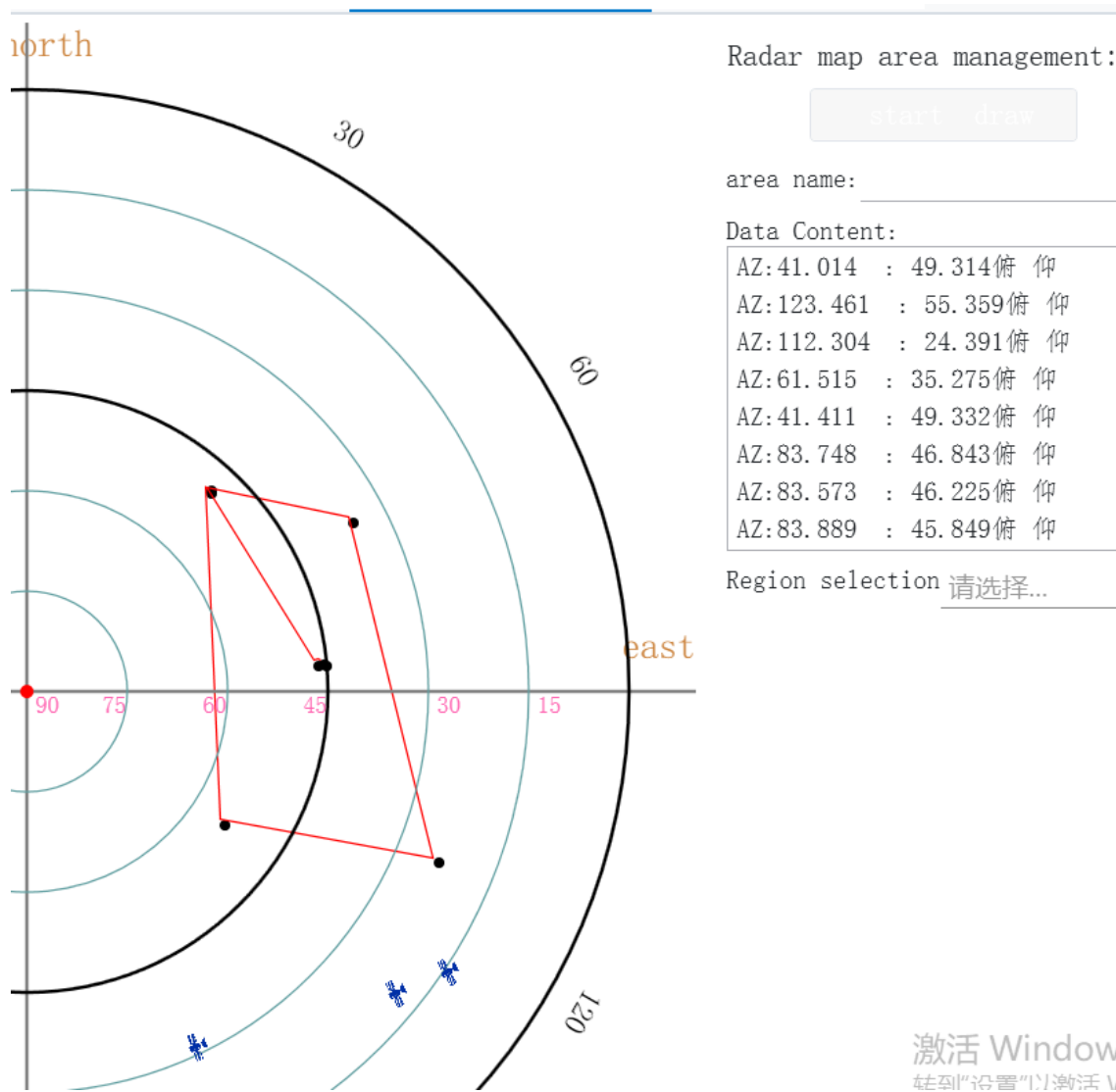


Figure 3.24

- Map area tracking: Draw the information of the trackable satellite target on the map for tracking. Mainly used for regional signal tracking. Draw the area on the flat map and save it. Select the saved area in the map area and perform tracking. When using this function, you have a deep understanding of the satellite signals in this area.

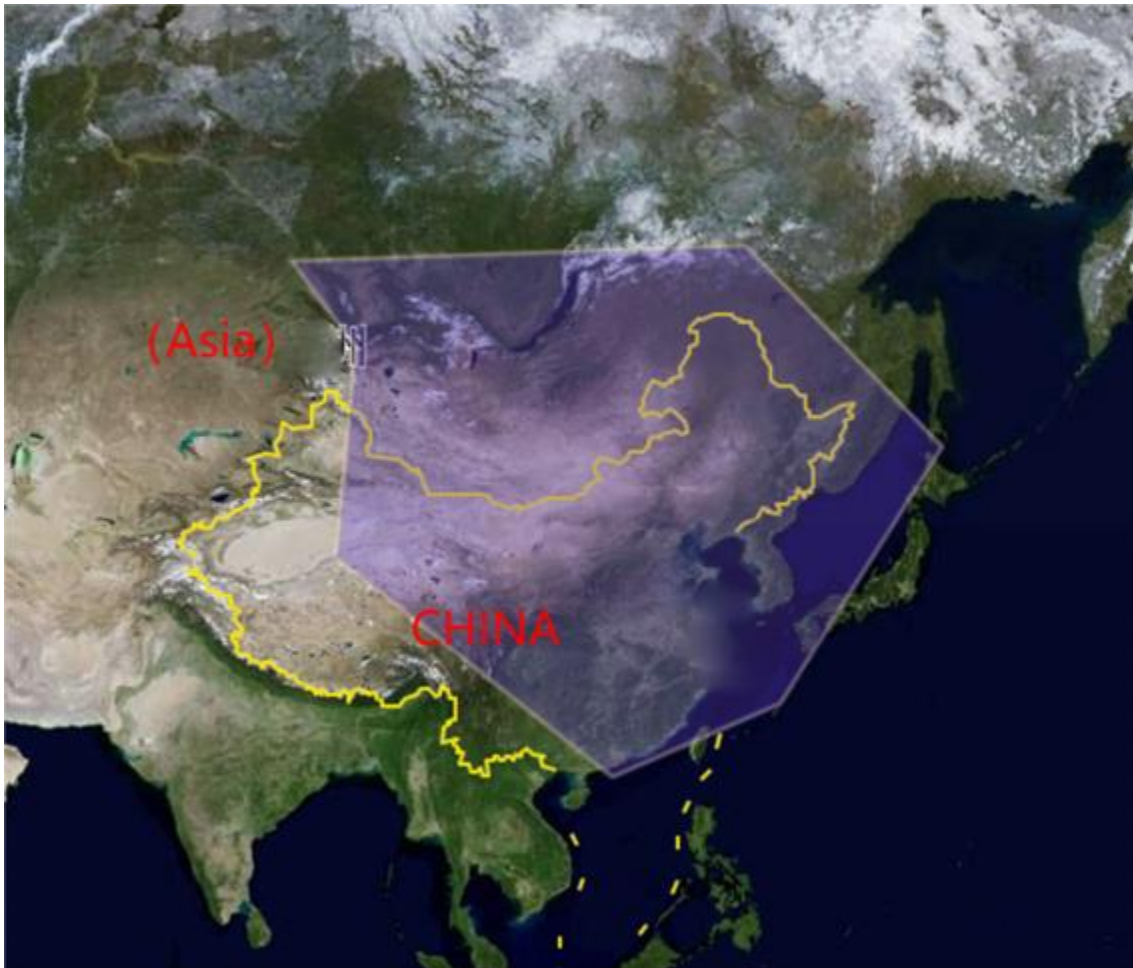


Figure 3.25

- File tracking: Import the azimuth and elevation of the tracking information for tracking. The home page is used for three-party data interface and external control tracking. The data format is "azimuth\tab elevation\tab polarization".

Tracking Settings:

1. Second granularity: Indicates how many times the system controls the antenna movement per second. Second granularity will affect the accuracy of the points generated by the software during the tracking process. It is recommended not to exceed "10". The larger the granularity, the better. When the granularity becomes larger, it may cause the software to occupy a large amount of system resources and network resources. It is recommended to use "2" for low-orbit satellites and "1" for high-orbit and medium-orbit satellites. After clicking Set, it will be used in the next calculation process.
2. Global tracking minimum elevation limit: limit the elevation angle of the antenna during tracking, the lowest angle cannot be lower than this angle. Generally set to 10. Below 10 degrees, the signal is generally poor and the obstruction is obvious. Click to set to take effect.

3. Time offset: used to adjust the current system time offset to act on local time and antenna time, in seconds, and can be positive or negative. The modification takes effect immediately.
4. Azimuth offset: used to modify the offset scale in azimuth, and the modification takes effect immediately. For antennas that can be automatically calibrated, this parameter is generally modified within the antenna's attitude compensation.
5. elevation offset: used to modify the offset scale on the elevation, and the modification takes effect immediately. For antennas that can be automatically calibrated, this parameter is generally modified within the antenna's attitude compensation.

task Settings

Second granularity: times per sec

Minimum Tracking Elevation Limit: °

[Settings](#)

Time Offset: S

Azimuth offset: °

Elevation Offset: °

AGCAutomatic adjustment: ☐

amplitude V

KValue

Figure 3.26

4. Daily Maintenance

Carry out daily inspections on the equipment and the technical condition of each part of the equipment to promptly identify hidden dangers and take countermeasures to ensure the normal operation of the antenna. For example, check whether the indicator light on the antenna power box panel is working properly and whether the azimuth and elevation movements are normal. Before use, check whether the equipment is correctly connected to other equipment. After use, check whether the switch is in the off state and whether the buttons are restored to the preset state. Check the wires and cables. If they are broken, aged, or the connectors are damaged, repair or replace them in time.

4.1 Equipment Maintenance

- a) Pay attention to maintaining a good environment;
- b) The rotating parts of the antenna need to be regularly and quantitatively refueled (greased) and replaced with oil (grease) to ensure that the quality of the lubricating oil or grease meets the requirements, the oil gun, oil cup, and oil pot are complete, and the oil path is unobstructed;
- c) Clean the antenna control box (electrician cooperation is required) to ensure that the lines are fixed neatly and the safety protection is reliable ;
- d) Check whether the outer surface of the equipment is rusted or damaged, and apply oil and repaint if necessary;
- e) Check whether the modules of each functional unit of the equipment are loose, wipe and remove all kinds of stains on the contact points of the connectors, and check whether the connecting wires inside the equipment chassis are loose. If the equipment has been used for more than two years,

- check all cables and replace them if necessary;
- f) Check whether the shells of electronic components on the printed circuit board are deformed, leads are loose or disconnected, pins are rusted, burned, oxidized, etc., and perform replacement inspection on spare parts of the equipment;
 - g) When cleaning the equipment, use a detergent with strong detergency and insulating properties and a soft wipe, and be gentle when operating;
 - h) Implement a system of designated personnel, designated machines, credentialed operation and shift handovers, be familiar with the equipment structure, abide by operating procedures, use it reasonably, maintain it carefully, and ensure safety without accidents.

5. Troubleshooting

This section introduces how to quickly analyze and determine the fault module for reference by maintenance personnel.

Note: If the antenna does not work properly, first make sure that the external connection cable is well connected, without disconnection or short circuit, and then proceed to find the fault point. Before checking the internal module, also make sure that the internal connection cable is well connected, without disconnection or short circuit, and then further check the module fault. Avoid module inspection errors caused by cable faults.

5.1 Common hardware fault diagnosis

5.1.1 Network module troubleshooting

The main reason for the network module failure is that the network cannot be connected. Turn off the power, open the upper cover of the control box, check all internal connection cables to ensure that the connection is good, and then restart the machine. If the fault persists, replace the original board with a good network module and restart the machine. If the fault is eliminated, it can be determined that the network module is faulty, otherwise it is other modules that are faulty.

The network fault diagnosis flowchart is shown in the figure below.

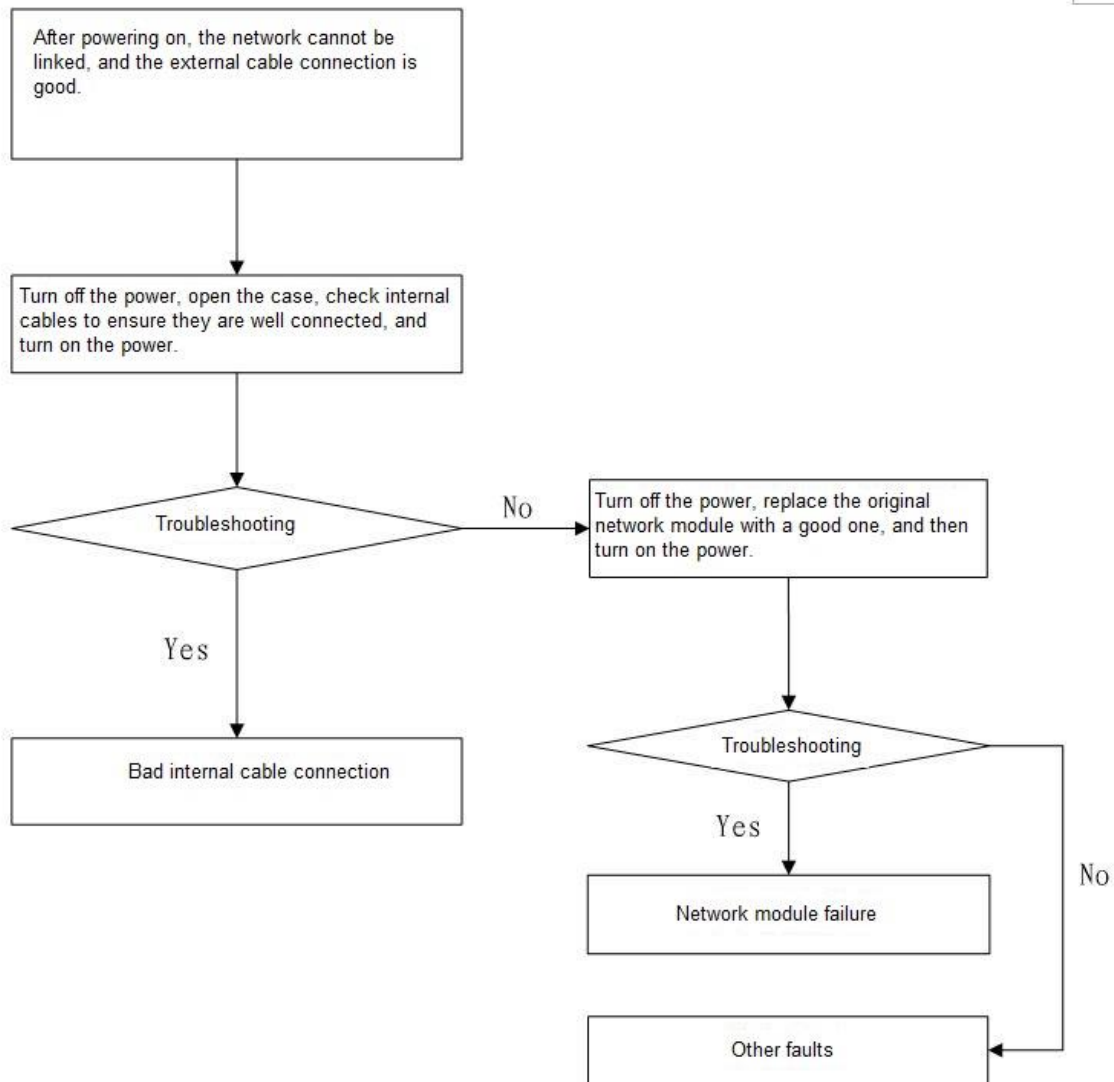


Figure 5.1

5.1.2 Control board module fault diagnosis

The control mainboard module mainly provides the encoder interface, limit information interface, COM interface, communication interface with the monitoring board, and CAN bus channel of the antenna controller. If the alarm light comes on after powering on, the alarm information interface shows abnormal communication, multiple ports cannot communicate with the outside, the antenna movement angle is abnormal, and the working indicator light is abnormal, it may be a fault in the control mainboard module. After turning off the power, open the upper cover of the control box, check all internal connection cables to ensure that the connection is good, and then restart the machine. If the fault is not eliminated, replace the original board with a good control mainboard module, and then restart the machine. If the fault is eliminated, it can be determined that the control mainboard module is faulty, otherwise it is a fault of other modules.

The control main board module fault diagnosis flow chart is shown in the figure below.

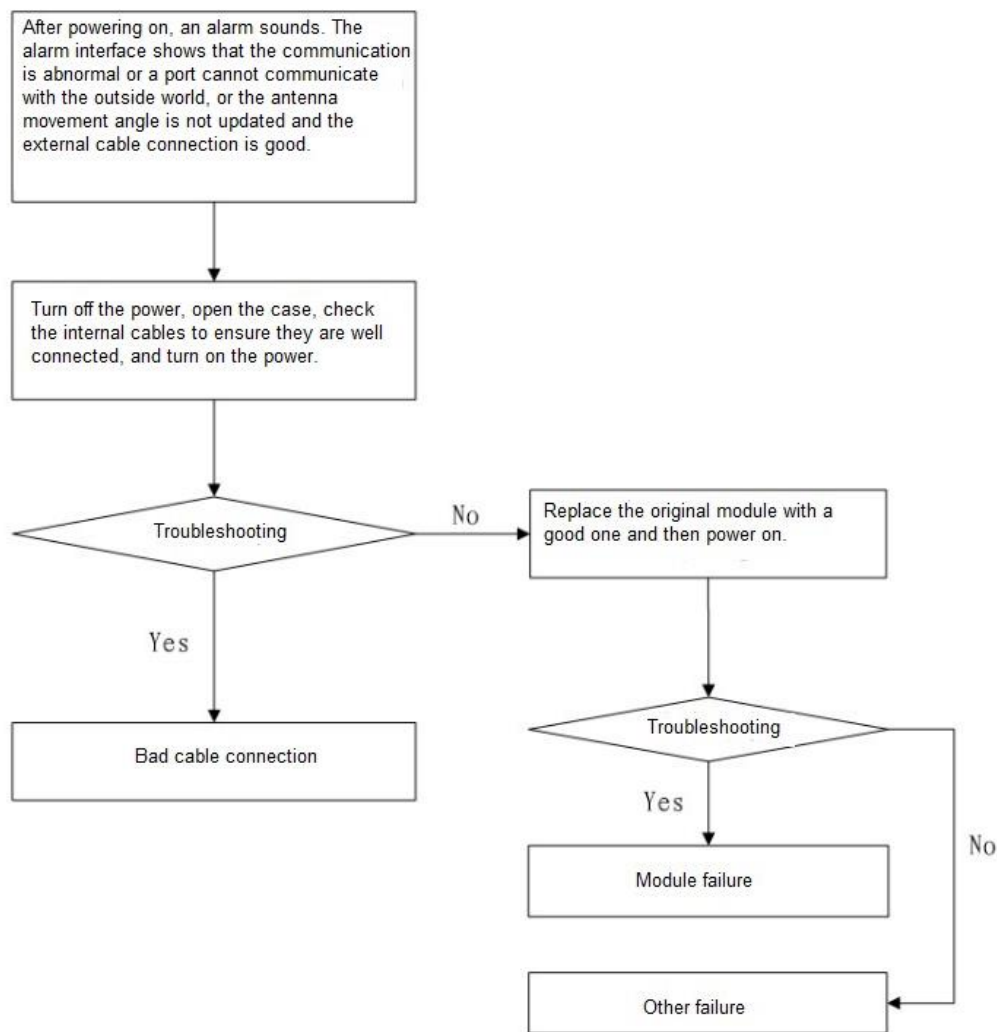


Figure 5.2

5.1.3 Power module fault diagnosis

After powering on, if there is no response, you need to turn off the power and open the cover of the power box to repair the fault. Remove the connection cable of the faulty power module, turn on the power again, and use the voltage block of the multimeter to measure whether the voltage between -V and +V of the power module is correct. If it is correct, it is a fault of other modules, otherwise it is a fault of the power module.

It should be noted that if the power module has no voltage output, you should first check whether there is a ~220V voltage input. If so, check whether the fuse on the module is blown. If not, check the power input of the power box.

5.1.4 Driver module fault diagnosis

The driver module mainly drives the motor to rotate. If the alarm indicator light comes on after the machine is turned on, the alarm information interface shows that the polarization drive is abnormal, the elevation drive is abnormal, or the azimuth drive is abnormal, or the motor does not rotate, the motor speed is abnormal, the motor cannot stop rotating, etc., it may be caused by the driver abnormality. After turning off

the power, check all the connecting cables related to the driver and motor of the equipment to ensure that the connection is good, and then restart the machine. If the fault is not eliminated, replace the original module with a good driver module, and then restart the machine. If the fault is eliminated, it can be determined that the driver module is faulty, otherwise it is other modules.

The driver module fault diagnosis flow chart is shown in the figure below.

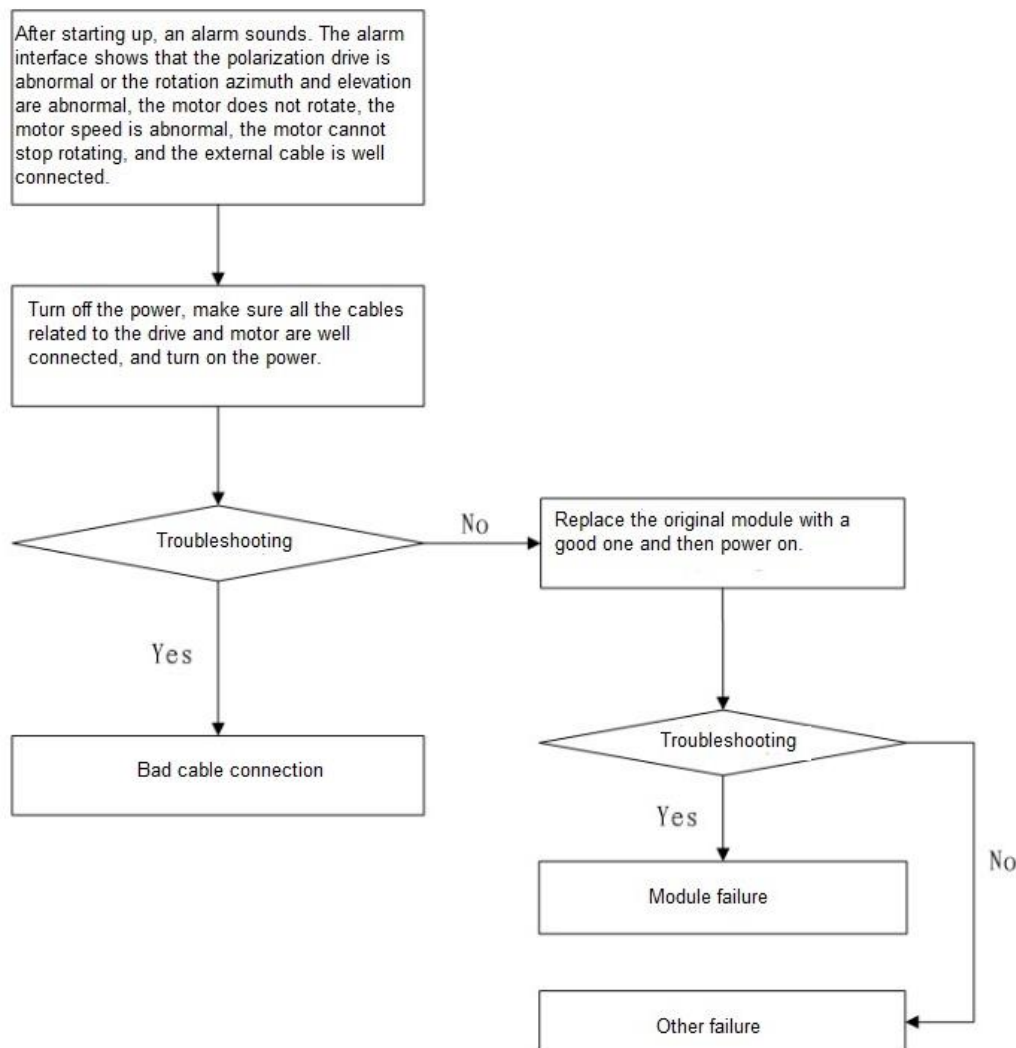


Figure 5.3

5.1.5 Motor fault diagnosis

The main manifestation of motor failure is that the operating antenna rotates, but the motor does not rotate or rotates with abnormal noise or abnormal speed. After turning off the power, check all the connecting cables related to the driver and motor of the equipment to ensure that the connection is good, then restart the machine. If the fault is not eliminated, replace the original module with a good driver module, and then restart the machine. If the fault is eliminated, it can be determined that the motor is faulty, otherwise it is other module faults.

Motor fault diagnosis flow chart.

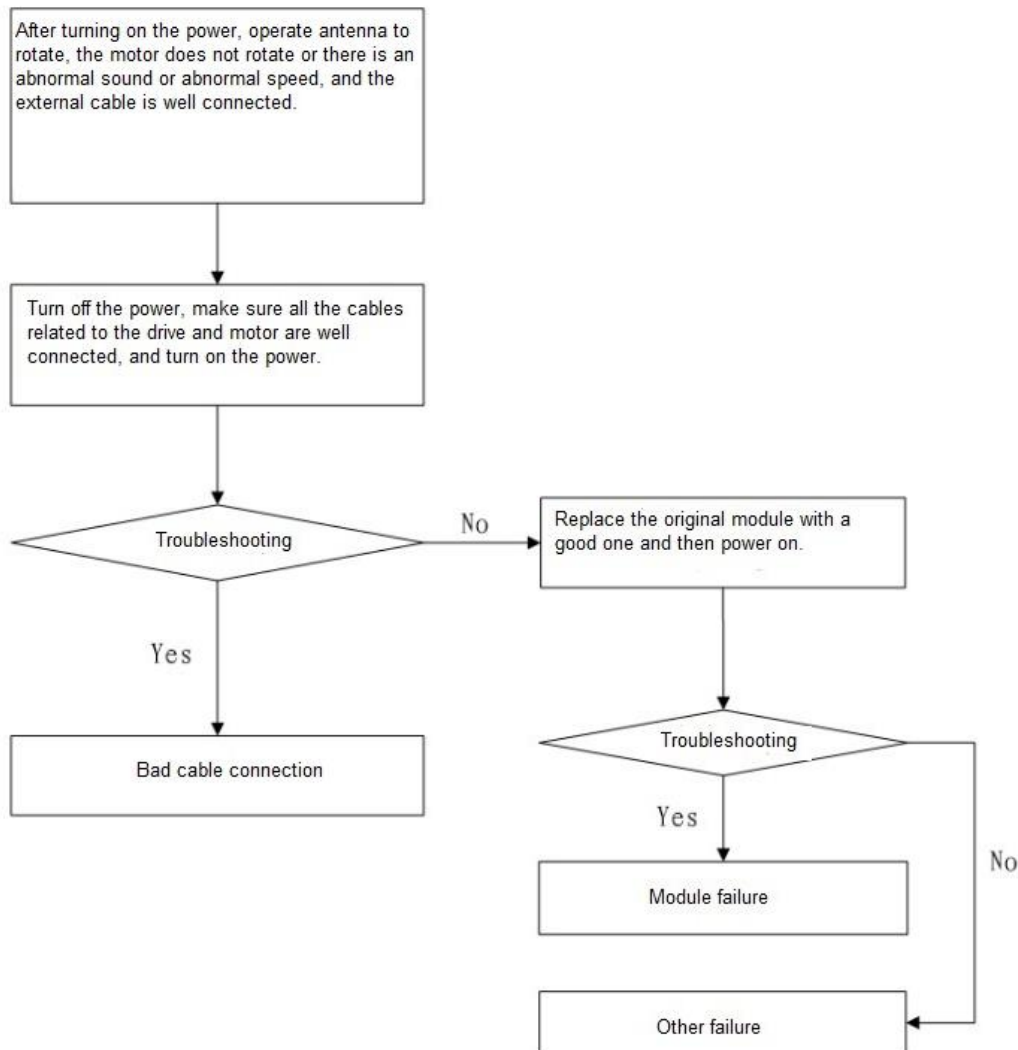


Figure 5.4

5.2 Common software troubleshooting

The software has an alarm function for faults. When the antenna is abnormal, the antenna is prohibited from running, and the reason for the stop can be found in the alarm status bar. Common software faults are as follows:

5.2.1 Azimuth drive abnormality

The azimuth drive wiring is abnormal or the azimuth drive is faulty. First, make sure that the azimuth drive connection cable is well connected, without disconnection or short circuit, and then start to find the fault point. Before checking the internal module, make sure that the internal connection cable is well connected, without disconnection or short circuit, and then further check the module fault according to the figure below.

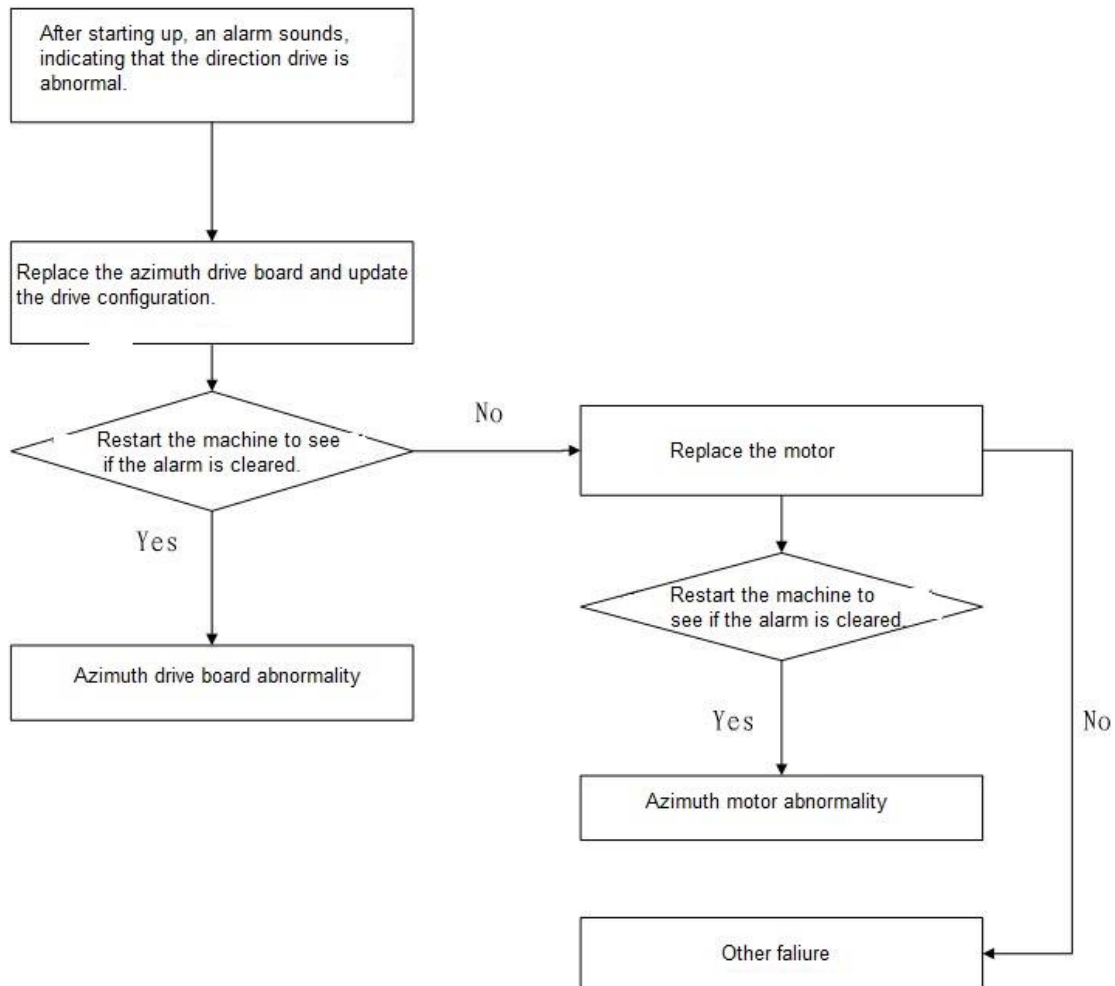


Figure 5.5

5.2.2 elevation drive abnormality

The elevation drive wiring is abnormal or the elevation drive is faulty. First, make sure that the elevation drive connection cable is well connected, without disconnection or short circuit, and then start to find the fault point. Before checking the internal module, make sure that the internal connection cable is well connected, without disconnection or short circuit. The troubleshooting method is as shown in the figure below.

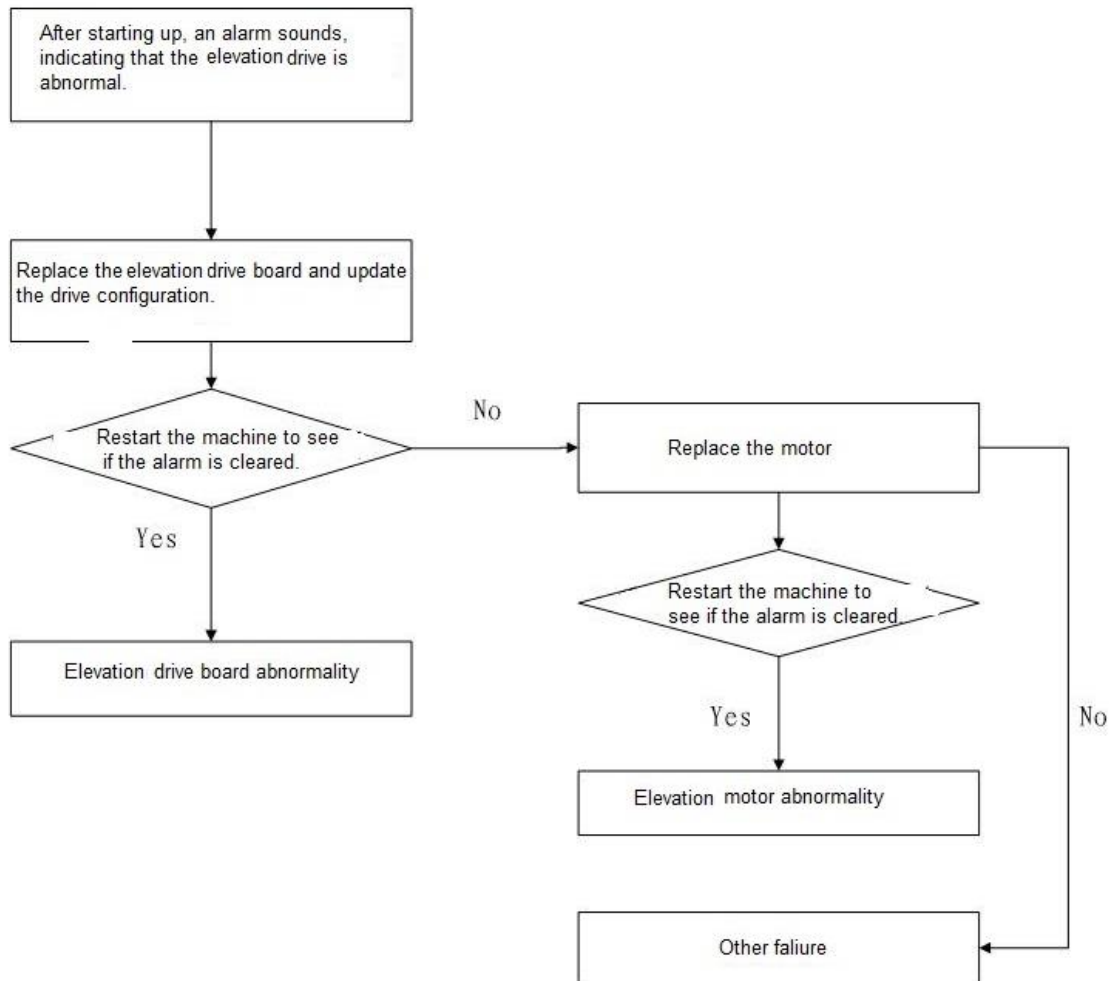


Figure 5.6

5.2.3 CAN communication abnormality

The STM32 is communicating abnormally with the azimuth or elevation driver. First, make sure that the cables connecting the control box and the antenna motor are well connected, without disconnection or short circuit, and then start to find the fault point. Before checking the internal module, make sure that the internal connection cables are well connected, without disconnection or short circuit, and check according to the figure below.

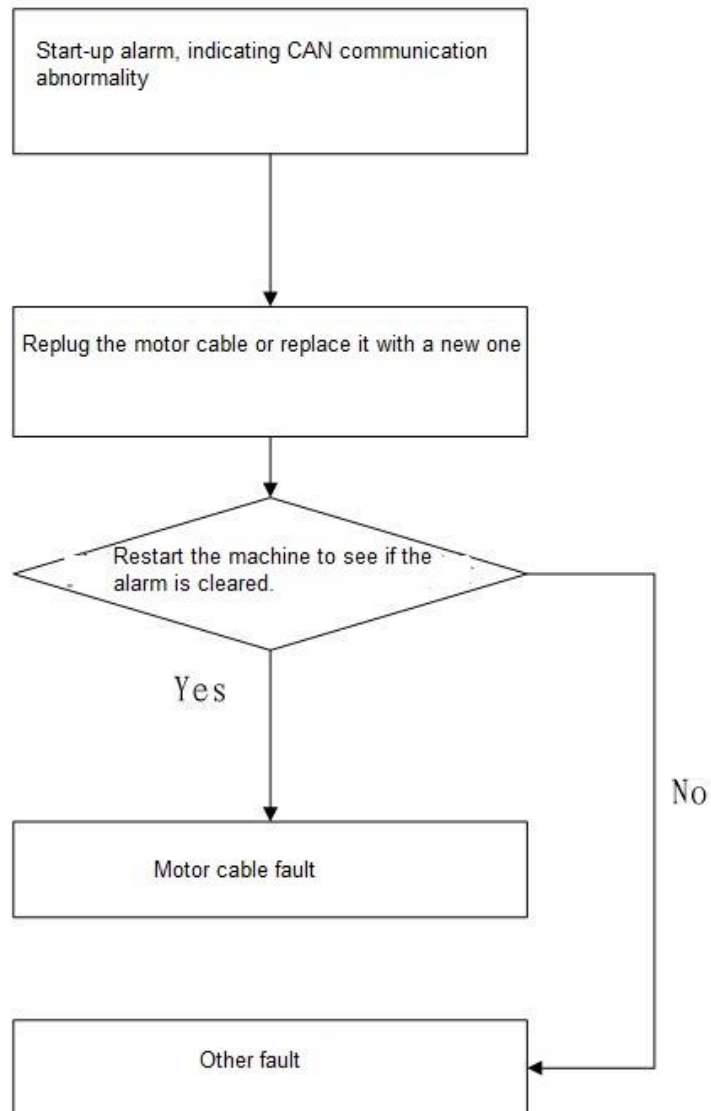


Figure 5.7

6. Precautions

6.1 Precautions for daily use

- a) Before powering on for the first time, use a multimeter to check whether the 220V AC power supply is normal. Also check whether the external cables are connected correctly and reliably. A power supply with a grounding wire must be used for power supply.
- b) If you encounter an unexpected situation where the antenna is out of control during use, please immediately press the power switch button on the power box.

6.2 Shutdown precautions

Before turning off the power switch, wait until the antenna switch button goes out before turning off (or disconnecting) the external input power, otherwise the operating system may be damaged!

6.3 Other considerations

- a) Pay attention to maintaining a good environment;
- b) When encountering strong winds exceeding the working wind speed, the antenna should be stored in time and the feed source should be removed;
- c) Check the antenna fasteners and rust conditions regularly once a year, and replace the fasteners that are severely corroded (threads are almost indistinguishable) in time; apply calcium-based grease (butter) to the lead screw and elevation axis screws for protection;
- d) When cleaning the equipment, clean the sand and dust from the moving joints from time to time. Use non-corrosive cleaning agents and soft wipes, and be gentle when operating.
- e) The reducer should be greased and replaced in time;
- f) Check whether the shells of electronic components on the printed circuit board are deformed, leads are loose or disconnected, pins are rusted, burned, oxidized, etc., and perform replacement inspection on spare parts of the equipment;
- g) Clean the antenna control box (electrician cooperation is required) to ensure that the lines are fixed neatly and the safety protection is reliable;
- h) Implement a system of designated personnel, designated machines, credentialed operation and shift handovers, be familiar with the equipment structure, abide by operating procedures, use it reasonably, maintain it carefully, and ensure safety without accidents.



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